

Evolutionary Computing (X\_400111)

Standard Assignment, Task II: Neuroevolution

### **Send the Best, the Worst, or Anyone?**

Emigrant Selection in Island-Model Neuroevolution  
of a Walking Robot

Team 92

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# Send the Best, the Worst, or Anyone? Emigrant Selection in Island-Model Neuroevolution of a Walking Robot

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## ABSTRACT

We evolve the 212 weights of a 16-8-4-8 neural network that makes the eight-hinge spider\_8 robot of the ARIEL framework walk 2 m to a target on the OlympicArena within 15 s. Our research question is how the emigrant-selection policy of a four-island ring model (migrating the best, the worst or random individuals, or not migrating at all) affects the convergence speed and the final quality of evolution. In 20 paired runs of 12,000 evaluations per condition, every evolutionary algorithm (EA) clearly outperforms random search (final fitness 1.13–1.24 against 2.49, lower is better; Holm-corrected paired  $t$ -tests,  $p < 10^{-9}$ ). Between the policies the differences are small: migrating the best or random individuals probably converges faster than no migration (posterior probabilities 99% and 98% on the area under the convergence curve), but after correction for 15 pairwise comparisons no difference between two EAs is significant in speed, final fitness or distance on 20 unseen arenas. Migrating the best made the islands as alike as a single population, yet this cost little within 166 generations; migrating the worst behaved like no migration. Random migration ends with the best mean fitness and is never the worst EA on any seed.

## KEYWORDS

Neuroevolution, island model, migration policy, evolutionary robotics, legged locomotion

## 1 INTRODUCTION

Evolutionary robotics designs robot controllers by evolving them in simulation [18]. A common choice is to fix the topology of a neural network and evolve its weights [10]: a continuous search space of hundreds of dimensions, rugged and multimodal, in which a population can converge prematurely on a mediocre gait.

Island models are a standard defence against such premature convergence. The population is split into sub-populations (islands, or demes) that evolve separately and periodically exchange a few individuals (migration) [1, 3, 23]. Isolation lets the islands explore different regions; migration spreads good genetic material. How an island model behaves depends on the topology, the interval and size of migration, how emigrants are selected and whom they replace [21]. Cantú-Paz [4] showed that choosing emigrants by fitness raises the selection pressure of the whole algorithm, which speeds up convergence but risks driving every island to the same solution. These results concern genetic algorithms on benchmark functions. Neuroevolution adds a complication: two networks that solve a task can order their hidden neurons differently (the competing-conventions problem [19]), so recombining

a native with an immigrant from another lineage can produce a non-functional child. Whether migrating the best is still the right choice for evolving a walking robot is therefore an open, practical question.

We study this question on the assignment’s task: evolving the weights of a neural-network controller so that a fixed robot body walks from its spawn point to a target. Our research question is:

**RQ.** How does the emigrant-selection policy of an island-model evolutionary algorithm (EA) (migrating the *best*, the *worst* or *random* individuals, or *no* migration) affect the convergence speed and the final quality of evolved locomotion controllers?

Following Cantú-Paz [4], we expect: **H1**, every migration policy converges faster than no migration; **H2**, migrating the best converges fastest but ends at an equal or worse plateau than random migration, because the islands lose their differences; **H3**, migrating the worst behaves close to no migration. We compare the four policies with a standard single-population EA and random search on 20 paired seeds each, with a migration interval chosen in a preliminary study, and then explore which algorithm ends best.

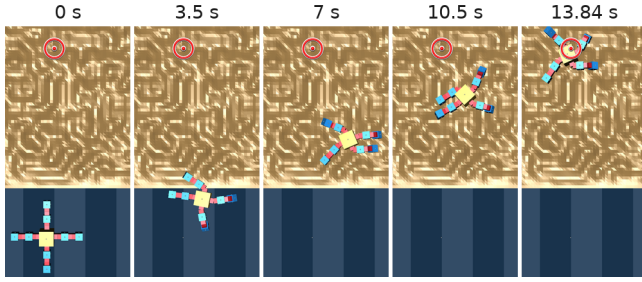
## 2 METHODS

### 2.1 Task, body and world

The body is spider\_8 from ARIEL’s John Set: a core with four legs of two hinges each, eight position-controlled motors in total. The world is the OlympicArena: a flat start, then from  $x = 0.5$  m a strip of centimetre-high bumps on which the target lies at (2, 0) m (Figure 1). The robot starts at the origin, and each walk lasts 15 s, as in the assignment’s template; it ends early when the core comes within 0.1 m of the target. The network is queried at 50 Hz (every 10 physics steps of 2 ms); the motors’ servos move the joints in between.

Body and world were free choices, made in pilot runs and then fixed. Our first body, spider\_16, cannot lift itself: its 4 kg need about 5 N m per leg, but every John Set motor delivers at most 0.66 N m. On ARIEL’s rugged world (mean slope  $25^\circ$  along the path) none of our bodies got far; on the flat world spider\_8 solves the task within about 2,000 evaluations, leaving no room for conditions to differ. The OlympicArena lies in between: learnable but not trivial.

ARIEL places the bumps at random whenever the arena is built, so we build and save one arena per seed, and every condition with that seed walks it. This makes the fitness deterministic on a given machine and pairs the conditions by seed. Generalisation is tested afterwards on 20 unseen arenas (Section 3).



**Figure 1:** The best controller of one standard-EA run (seed 25) walking on its training arena, seen from above. It starts on the flat part (bottom), crosses onto the bumpy strip at  $x = 0.5$  m and arrives, its core within 0.1 m of the target (red circle), after 13.84 s.

**Table 1:** The controller’s 16 inputs, all scaled to about  $[-1, 1]$ .

| Input        | Size | Meaning                                 |
|--------------|------|-----------------------------------------|
| Joint angles | 8    | current angle of each hinge             |
| Clock        | 2    | sin and cos of a 1 Hz beat, for rhythm  |
| Target       | 3    | distance / 2 m; sin, cos of its bearing |
| Tilt         | 3    | the up direction in the core’s frame    |

## 2.2 Controller

The controller is a fully connected feed-forward network with two hidden layers of 8 and 4 tanh neurons and 8 tanh outputs, each scaled to a target angle in  $[-\pi/2, \pi/2]$  for one hinge. With a bias on every neuron, it has 212 weights, which form the genome. Its 16 inputs are listed in Table 1. As the network has no memory, a clock input supplies the rhythm; it is a fixed signal rather than an oscillator inside the controller, so the controller is not a central pattern generator (CPG), which the assignment excludes. The target is given relative to the robot’s heading rather than as an absolute position, so that “the goal is 1 m away,  $30^\circ$  to the left” means the same everywhere on the arena.

*Inputs and network shape.* Our first controller also sensed the terrain with distance rays, made for the rugged world’s hills. On the OlympicArena a ray reads up to 3 m and the bumps are about  $\pm 2$  cm, so a bump moves a ray input by only about 0.01, and without rays the EA trained as well (Table 2); we removed vision. The 16-8-4-8 shape won a comparison of six (Table 2), plausibly because four shared signals push the eight legs towards coordinated patterns.

## 2.3 Fitness function

The assignment’s fitness is the planar distance  $d_T$  between the core’s final position and the target. Early in the study we added penalties for the share of the walk in which the core touches the ground or the robot is upside down, as a robot can otherwise drag its core towards the target. The best controller of a long pilot run with this fitness came within 0.84 m of the target, but in replays it crouched (the core 1.6 cm above lying height, although the body

can hold it 6 cm up) and kept one leg tucked away ( $5\text{--}9^\circ$  of commanded movement against  $17\text{--}77^\circ$  for the other legs). The distance does not reward the mechanics of walking until they already produce progress, so we added terms for them and minimise

$$f = d_T + 0.5 \bar{d} + 1.0 c + 1.0 \ell + 1.0 u + 0.5 w, \quad (1)$$

where every term is measured at each 50 Hz update and, except for the two distances, lies in  $[0, 1]$ :

- **Be fast** ( $\bar{d}$ ): the distance to the target averaged over the whole 15 s, counting 0 after arrival, so that closing the distance early and arriving early both lower the fitness.
- **Stand** ( $c, \ell$ ):  $c$  is the share of the walk in which the core touches the ground, and  $\ell$  is the mean shortfall of the core’s height  $h$  above its lying height from 4 cm,  $\min(1, \max(0, 1 - h/4 \text{ cm}))$ . The body can hold itself 6 cm up, so 4 cm is demanding but reachable, and every millimetre counts.
- **Stay upright** ( $u$ ): the share of the walk spent upside down.
- **Use all legs alike** ( $w$ ): each leg should do about a quarter of the work. With  $W_i$  the positive motor work of leg  $i$  and  $\bar{W}$  its mean over the legs,  $w = 1 - \min_i W_i / \bar{W}$ : 0 when every leg works alike, 1 when one leg is idle. Measuring how much each leg moves missed a leg that moved without pushing; only the motor work exposed it (22 J against up to 40 J for the other legs).

The term weights were set by hand in pilots and not tuned; with weight 1, spending the whole walk on the ground costs as much as ending 1 m further away. Rhythm is not a fitness term: the clock input provides it. A robot that holds still scores 5.0 ( $d_T = \bar{d} = 2$ , core on the ground, nothing lifted), and one that arrives at once while standing tall and using all legs alike scores close to 0. A typical evolved controller (no migration, seed 10) scores  $0.29 + 0.5 \cdot 1.12 + 0 + 0.09 + 0 + 0.5 \cdot 0.11 = 1.00$ : the two distance terms dominate, and the gait terms decide between walkers that get about equally far. This behavioural shaping prescribes part of *how* to walk, which the assignment allows for a clearly defined metric; all conditions share it, and we also report the plain distance.

## 2.4 The evolutionary algorithm

The genome is a fixed-length vector of real numbers, so we use a real-coded genetic algorithm: Gaussian mutation explores around good weights, and recombination can combine good neurons found in different individuals. It is built on ARIEL’s `ariel.ec` (population, generation loop and Gaussian mutation); selection, crossover and migration are our own.

*Representation and variation.* A genome is the vector of 212 real weights, initialised from  $\mathcal{N}(0, 0.5^2)$  as in the assignment’s template. Parents are chosen by tournaments of three within an island [9]. With probability 0.9 a child is made by *neuron-level uniform crossover*: for every hidden neuron, a fair coin decides from which parent the child inherits all of that neuron’s incoming weights and bias (the second hidden layer’s neurons also take their outgoing weights); each output bias comes from a random parent. Otherwise the child copies one parent. Every child is then mutated by adding  $\mathcal{N}(0, 0.05^2)$  noise to every weight. Keeping neurons whole is the classic response to the competing-conventions

problem [16, 17]: the child inherits working units rather than a mixture of two units that may play different roles.

*Survivor selection.* Each generation, an island keeps its two best individuals unchanged (elitism) and replaces the other 18 with children. As the arena is fixed, the fitness is deterministic and elites are not re-evaluated.

*Island model.* Four islands of 20 individuals form a ring. Every 20 generations each island copies two emigrants to the next island, where they replace that island’s two worst individuals. The emigrant-selection policy is the variable of the RQ: *best* (the two fittest), *worst* (the two least fit), *random* (two chosen uniformly) or *none* (no migration, four isolated islands). Everything else is identical across conditions.

*Baselines.* The *standard EA* is the same algorithm with one population of 80 and 8 elites, so only the population structure differs. *Random search* draws every new genome from the initial distribution and keeps the best, at the same budget, as the assignment requires.

*Budget.* Each run stops at the first generation that completes 12,000 evaluations: 80 initial evaluations, then 72 children in each of 166 generations, 12,032 in all. A plateau-based stopping rule never triggered within our pilot budgets, so we chose the budget from where the pilot curves flattened: 95% of the progress of a long run came within 5,000 evaluations and its last improvement at 8,800.

## 2.5 How the settings were chosen

Table 2 lists every setting with its justification. Settings were changed by controlled comparisons with a decision rule fixed beforehand: a new value replaces the current one only if its best fitness after 6,000 evaluations is better on all three tuning seeds and on the mean. The tuning seeds (100–102) and their arenas are disjoint from those of the final experiment (10–29), and tuning used the random emigrant policy, the middle ground between sending the best and the worst. Two tested values passed the rule: crossover probability 0.9 (mean best fitness 1.369 against 1.800 at 0.5) and the 16-8-4-8 network; vision was dropped on a tie (above). The numbers of islands and migrants, the topology and the replacement were not tuned: they set the context of the RQ and take common values from the literature [4]; the interval came from a preliminary study (Section 4.1).

## 3 EXPERIMENTAL SETUP

*Conditions.* We compare six conditions: the island EA with the *best*, *worst*, *random* and *none* policies, the standard EA and random search. Each runs on seeds 10–29, 120 runs in all. A seed fixes the EA’s random numbers, so all conditions start from the same 80 initial networks, and selects that seed’s saved arena; the runs are thus paired by seed. Twenty seeds allow a paired test of every pair of conditions even after correcting for 15 pairs; with five, the smallest possible two-sided Wilcoxon  $p$  is 0.0625.

*Preliminary study.* A first version of this experiment ran the same six conditions on five seeds (0–4), migrating every 10 generations, and then the *best* policy every 5, 20 and 50 generations

on the same seeds and arenas (Section 4.1). It set the interval of the final experiment to 20, a design choice rather than a tuning result: at 20 the islands stay distinct for longer, so the emigrant policy has more to act on. Because the interval was chosen on seeds 0–4, the final experiment uses 20 fresh seeds and arenas.

*Metrics.* Lower is better for every metric. (1) The best-so-far fitness against evaluations, and its area under the curve (AUC; the mean best-so-far fitness over the budget), which rewards being good early. (2) Convergence speed: the median number of evaluations until the best fitness first drops below 1.6, a run that never gets there counting as slowest. This threshold separated fast from slow runs in tuning: no run with the initial settings reached it within 6,000 evaluations. (3) The final best fitness, and the smallest final distance  $d_T$  of any walk of the run. (4) Generalisation: each run’s best controller walks 20 unseen arenas (the same 20 for every run) for 15 s, and we report the mean remaining distance; a robot that falls off an arena counts as 10 m away. (5) Diversity (descriptive): the mean distance of the 80 genotypes to their centroid at the end of the run, large when the islands have evolved apart.

*Statistics.* Every comparison is paired by seed. A Friedman test [6] across the six conditions asks whether they differ at all. Then every pair of conditions (15 pairs) is compared on the final fitness, the AUC and the unseen distance with a two-sided paired test, Holm-corrected across the pairs of each metric [12]. The choice of test was fixed before any comparison was possible (before the first of the 120 runs had finished): if no pair’s per-seed differences reject normality (Shapiro–Wilk, Holm-corrected,  $\alpha = 0.05$  [20]), the metric gets the paired  $t$ -test; otherwise the Wilcoxon signed-rank test [24], the usual choice for comparing evolutionary algorithms [6, 7], which needs no normality and loses little power when the data are normal [11]. We test normality rather than assume it because the arenas differ in difficulty: in the preliminary study the seed explained 37% of the EAs’ variance in final fitness and the condition 14%, so a hard arena can make one seed’s difference an outlier. Effect sizes are the mean paired difference with its 95% CI and  $d_z$ , that difference divided by the standard deviation of the per-seed differences (by Cohen’s convention,  $|d_z| \approx 0.5$  is a medium effect [14]). For every pair we also give the posterior probability that one condition has the lower true mean, from a Bayesian paired  $t$ -test with a non-informative prior [2]. It is not corrected for multiple comparisons, so it describes the evidence; only the Holm-corrected tests decide what we call significant.

*End results.* After seeing the results, we added an exploratory analysis of where the runs end, giving priority to reaching the target (Section 4.3). Every best controller also walks its own arena for 15, 20, 30 and 60 s and the unseen arenas for 30 s. Arrivals (yes or no) are compared with Cochran’s  $Q$  [5] and pairwise exact McNemar tests [15], continuous measures with the rule above; pairwise tests are Holm-corrected over the ten pairs of EAs.

*Compute and code.* The study took 345 runs, 3.62 M evaluations and 57 machine hours: 0.80 M evaluations to develop the task, body and fitness, 0.83 M for tuning, 0.54 M for the preliminary study and 1.44 M for the final experiment, whose 120 runs took 8.4 hours on two machines, split by seed (an Intel i7-12700H for seeds 10–19,

**Table 2: Every setting of the final EA and how it was chosen. Tests used the rule of Section 2.5 (3 seeds, 6,000 evaluations); numbers are mean best fitness (lower is better), “vs” the setting in use at the time, which changed as settings were adopted (early tests still had vision, one hidden layer of 16 and a stagnation rule). “Inconsistent”: not better on every seed.**

| Component             | Setting                                             | How it was chosen                                                                                                                                                      |
|-----------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network               | 16-8-4-8, tanh, 212 weights                         | tested against 16-8-8, 16-16-8, 16-32-8, 16-8-8-8, 16-16-16-8: the only shape better on every seed (1.203 vs 1.339 for 16-16-8)                                        |
| Inputs                | 16, no vision                                       | 10 → 5 rays (pilot); 3 and 1 ray worse (1.628, 1.513 vs 1.348); 0 rays tied (1.339), dropped                                                                           |
| Parent selection      | tournament, $k = 3$                                 | $k = 2$ and 5 tested: inconsistent (1.877, 1.612 vs 1.800)                                                                                                             |
| Crossover             | neuron-level uniform                                | tested against weight-level uniform (1.416), BLX-0.5 (1.168, better on 2 of 3 seeds) and headless chicken, i.e. crossover with a random genome (2.207) vs ours (1.203) |
| Crossover probability | 0.9                                                 | 0, 0.5 and 0.9 tested: 0.9 better on every seed (1.369 vs 1.800; 0 gave 2.063)                                                                                         |
| Mutation              | Gaussian, $\sigma = 0.05$ , every weight            | pilot over $\sigma \in \{0.05, 0.1, 0.2\}$ ; then 0.02, 0.1, and 10% of weights with $\sigma = 0.16$ tested: none better                                               |
| Mutation-step control | none                                                | a rule doubling $\sigma$ on stagnating islands tested: 1.369 vs 1.348 without, dropped                                                                                 |
| Survivor selection    | generational, 2 elites per island                   | 1 and 5 elites tested: inconsistent (1.791, 1.811 vs 1.800)                                                                                                            |
| Population            | 4 islands $\times$ 20                               | $4 \times 10$ and $4 \times 40$ tested: inconsistent (2.020, 1.693 vs 1.800)                                                                                           |
| Migration             | ring, every 20 generations, 2 copies, replace worst | literature [4], interval from the preliminary study (Section 4.1): every 20 kept the islands apart longest                                                             |
| Budget                | 12,032 evaluations (166 generations)                | pilot curves: 95% of the progress by 5,000, last gain at 8,800                                                                                                         |
| Walk                  | 15 s, ends at the target                            | the template’s length; 20 s training walks tested: no better on unseen arenas                                                                                          |

an Apple M3 Pro for 20–29). MuJoCo’s floating-point results differ slightly between these architectures and the differences grow over a walk, so a seed’s walks repeat exactly only on its kind of machine. Test runs, discarded runs and one showcase run are not counted. The code, which uses ARIEL (course fork, commit a435b19) unmodified, and one script per experiment are at <https://github.com/emre6943/ec-assignment2>; they and the saved arenas are also submitted with this report.

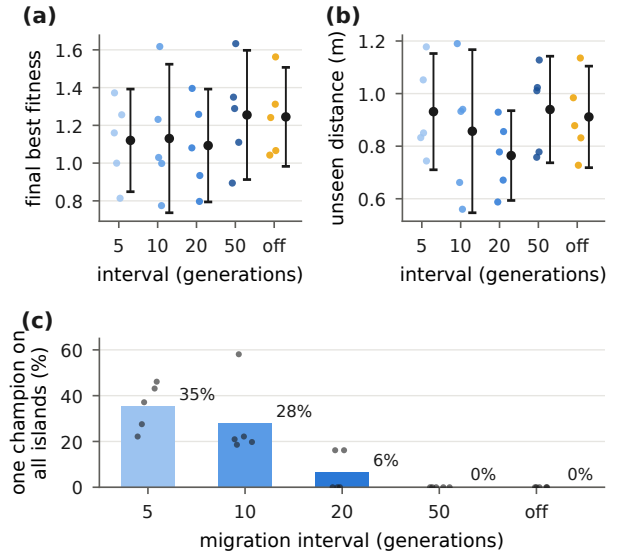
## 4 RESULTS

### 4.1 Preliminary study: how often to migrate

Migrating every 5, 10 or 20 generations was equally fast (AUC probabilities 42–58% between them) and faster than every 50 (93–99%) or never (96–100%); every 50 was only slightly faster than never (81%), as three migrations in 166 generations hardly matter. Of the fast intervals, every 20 kept one champion from taking all four islands longest (Figure 2c) and had the best final fitness (1.093) and unseen distance (0.76 m; Figure 2a–b), though only modestly (58% and 77% against every 10), so we chose it to keep the islands diverse.

### 4.2 Convergence and final fitness

Table 3 and Figures 3–4 summarise the final experiment. The conditions differ (Friedman  $\chi^2 = 46.6$  on the final fitness and 47.5 on the AUC, both  $p < 10^{-8}$ ), and the pairwise tests place the difference at random search: every EA outperforms it on every metric (Holm-corrected  $p \leq 0.007$ ; posterior probability  $> 99.9\%$ ). Random search never reached a fitness of 1.6 and came on average no closer than 1.21 m to the target (0.94 m at best). By the rule fixed in advance, the final fitness and the AUC got paired  $t$ -tests and the unseen distance Wilcoxon tests, because one no-migration controller fell off 3 of its 20 unseen arenas and skewed four of the five differences involving no migration.



**Figure 2: Preliminary study: the *best* policy migrating every 5, 10, 20 or 50 generations, or never (“off”), on seeds 0–4. (a) Final best fitness and (b) distance left on 20 unseen arenas: dots are seeds, black the mean with its 95% interval. (c) Share of generations in which all four islands hold the same champion; dots are seeds.**

Between the EAs the differences are small. By the AUC, migrating the best or random individuals is probably faster than no migration (Figure 3b; posterior 99% and 98%; best minus no migration  $-0.086$ , 95% CI  $[-0.152, -0.020]$ , effect size  $d_z = -0.61$ ,  $p = 0.014$ ), but neither difference survives the correction for 15 comparisons (Holm  $p = 0.14$  and  $0.27$ ). Migrating the worst is the slowest migration policy: best and random migration are faster with probability

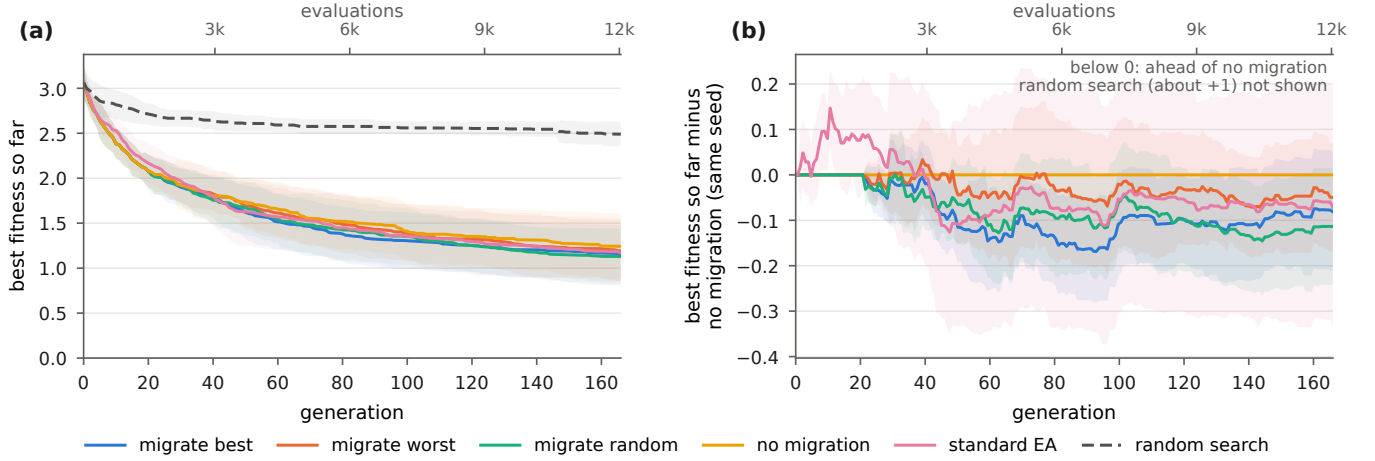


Figure 3: (a) Best-so-far fitness against generations (top axis: evaluations), mean over 20 seeds with a band of  $\pm 1$  standard deviation; lower is better. (b) Paired by seed: each EA’s best-so-far fitness minus that of no migration on the same seed, mean with 95% confidence band; below zero is ahead of no migration, and a curve’s mean over the run is the AUC difference. The island EAs equal no migration until the first migration (generation 20); random search (about +1) is left out of (b).

Table 3: Results of the final experiment at the end of the budget, mean  $\pm$  standard deviation over 20 seeds (lower is better). “Closest” is the smallest final distance  $d_T$  of any walk of the run; “to 1.6” is the median over all runs, one that never got there counting as slowest (runs that got there in brackets). P(faster than no migration) is the Bayesian posterior probability that the condition’s mean AUC is lower than without migration. Bold: the best EA per column.

| Condition      | Final fitness                       | Closest (m)                         | Evaluations to 1.6   | AUC                                 | Unseen arenas (m)                   | P(faster than no migration) |
|----------------|-------------------------------------|-------------------------------------|----------------------|-------------------------------------|-------------------------------------|-----------------------------|
| Migrate best   | $1.162 \pm 0.307$                   | $0.359 \pm 0.239$                   | 3,968 (18/20)        | <b><math>1.562 \pm 0.243</math></b> | $0.932 \pm 0.201$                   | 99%                         |
| Migrate worst  | $1.194 \pm 0.341$                   | $0.376 \pm 0.253$                   | 4,472 (17/20)        | $1.620 \pm 0.234$                   | $0.915 \pm 0.182$                   | 76%                         |
| Migrate random | <b><math>1.130 \pm 0.309</math></b> | <b><math>0.335 \pm 0.234</math></b> | 4,184 (18/20)        | $1.570 \pm 0.244$                   | $0.850 \pm 0.194$                   | 98%                         |
| No migration   | $1.243 \pm 0.365$                   | $0.402 \pm 0.259$                   | 4,976 (15/20)        | $1.648 \pm 0.252$                   | $1.005 \pm 0.338$                   | –                           |
| Standard EA    | $1.171 \pm 0.383$                   | $0.369 \pm 0.287$                   | <b>3,680 (17/20)</b> | $1.610 \pm 0.289$                   | <b><math>0.825 \pm 0.221</math></b> | 66%                         |
| Random search  | $2.490 \pm 0.135$                   | $1.208 \pm 0.096$                   | never (0/20)         | $2.612 \pm 0.068$                   | $1.406 \pm 0.118$                   | <0.1%                       |

90% and 94%, and it is faster than no migration with probability only 76%. Best and random migration reached a fitness of 1.6 in 18 of 20 runs, against 15 without migration, and sooner (median 3,968 and 4,184 evaluations against 4,976); the standard EA was quickest (3,680). In final fitness, random migration ended best (1.130; better than no migration with probability 96%, than migrating the best with 73%), and the standard EA lay among the island EAs (1.171). No paired test between two EAs is significant on any metric (final fitness: all Holm  $p \geq 0.81$ ).

*Diversity.* Migrating the best made the islands alike: at the end, its genotypes lay 1.68 from their centroid on average, closer than in the single population of the standard EA (1.93) and far closer than without migration (7.38, larger on all 20 seeds), whose islands end as far apart as random genomes (7.23 in random search). Random migration lay between (4.19), and migrating the worst kept the islands as far apart as no migration (7.12).

### 4.3 Exploratory: which algorithm ends best?

After seeing the results, we asked whether the EAs differ in where the runs end, giving priority to reaching the target (Table 4); as

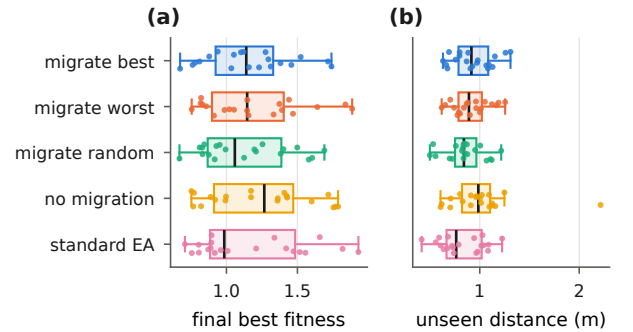


Figure 4: Every seed’s final best fitness (a) and distance left on the 20 unseen arenas (b), per condition: boxes show the quartiles and the median, dots are single seeds. The no-migration outlier in (b) fell off 3 arenas, each counted as 10 m. Random search (fitness about 2.5, 1.4 m unseen) is omitted for scale.

**Table 4: End results of the five EAs (exploratory). Own target: best controllers (of 20) reaching their training target within 15 and 30 s. Unseen arenas: share of the 400 walks reaching the target within 30 s (5 of all 2,000 do within 15 s), mean fitness over 15 s, and walks that fell off the arena within 30 s.**

| Condition      | Own target |           | Unseen arenas |              |          |
|----------------|------------|-----------|---------------|--------------|----------|
|                | 15 s       | 30 s      | reached       | fitness      | fell off |
| Migrate best   | <b>5</b>   | <b>10</b> | 6.0%          | 1.886        | 17       |
| Migrate worst  | 1          | 6         | 6.5%          | 1.854        | 7        |
| Migrate random | 4          | 7         | 10.2%         | 1.780        | 7        |
| No migration   | 4          | 7         | 7.8%          | 2.010        | 6        |
| Standard EA    | <b>5</b>   | 9         | <b>12.0%</b>  | <b>1.739</b> | <b>1</b> |

these measures were chosen post hoc, they describe rather than confirm. No end result separates the EAs significantly: their arrival rates on their own arena do not differ (Cochran’s  $Q$ ,  $p = 0.41$  at 15 s and 0.63 at 30 s), nor do their unseen arrivals or unseen fitness (Wilcoxon, all Holm  $p \geq 0.36$ ). Read descriptively, the standard EA generalised best: it reached the most unseen targets (12% of its walks against 6–10%), had the best unseen fitness and fell off an arena once in 400 walks, against 6–17 times for the island EAs. Random migration was the steadiest: in final fitness it was never the worst of the five on any seed, whereas the standard EA was the best on 9 seeds and the worst on 7 (Figure 4a). Migrating the best brought the most controllers to their own target within 30 s (10 of 20) but reached the fewest unseen targets and fell off most often. All controllers specialise to their arena: the best ones end on average 0.35–0.43 m from their own target after 15 s, against 0.83–1.01 m on unseen arenas. Finally, the 15 s limit is restrictive: 19 of the 120 best controllers reach their own target within 15 s, 39 within 20 s and only one more within 60 s; most others stall 0.1–1.0 m away.

## 5 DISCUSSION

*Answer to the RQ.* With 20 seeds, the emigrant-selection policy has at most a small effect on how fast the islands improve and none we can detect on where they end. H1 is supported only in direction: migrating the best or random individuals is probably faster than no migration (a medium effect for the best,  $d_z = -0.61$ ), but not significantly after correction. For migrating the worst, H3 holds rather than H1: it was the slowest migration policy, not clearly faster than no migration (76%,  $d_z = -0.16$ ), and its islands stayed as far apart as isolated ones. H2 is supported in direction: migrating the best was the fastest by the AUC, though tied with random migration (59%), and ended slightly behind it (73%, not significant); as most runs were still improving at the end, the plateau that H2 predicts was not reached. Its mechanism did appear [4]: the islands rarely shared one champion (4.3% of the generations; never with other policies), but their genotypes converged as closely as a single population’s (Section 4.2).

*Unexpected: the policies differ little.* Migrating the worst behaved like no migration, as H3 predicts: emigrants that lose most tournaments apparently pass on few genes, as the islands stayed apart. The surprise is that migrating the best changed the population so

much and its fitness so little: its lead over no migration shrinks from the AUC ( $d_z = -0.61$ ) to the final fitness ( $-0.29$ ), so within 166 generations migration mainly spreads good solutions sooner, and the lost diversity has not yet cost much on the training arena, although this policy reached the fewest unseen targets (Section 4.3). Nor did the competing-conventions risk of Section 1 show: the best immigrants are recombined with natives most often, yet that policy was among the fastest.

*Why the standard EA did as well.* It started slower (Figure 3b; behind no migration over the first 20 generations on 14 of 20 seeds): until the first migration the island EAs are four separate searches, and the best of four beat the single population of 80, although that population beat a typical island (50 of 80 islands at generation 10). Afterwards its best individuals breed with all 80 rather than within an island of 20, so good solutions spread at once; it caught up by generation 40. It was as good as the island EAs on average but less dependable: it either finds a good gait or gets stuck on its arena, while islands hedge against getting stuck, the classic case for island models. Its lead in the preliminary study (0.957 against 1.13–1.24) disappeared with 20 seeds, a reminder of how wide five-seed intervals are.

*Limitations.* The policy effects we found are small ( $|d_z| \leq 0.61$ ) and none survives the correction for 15 comparisons: our power to detect  $d_z = 0.6$  at the corrected level was about 30%, and 80% would need about 44 seeds. The settings were tuned one factor at a time on three seeds, which misses interactions [8]; tuning used the random policy and the interval study the best policy, so the settings may suit the policies unevenly and favour these two. Choosing the test by a normality check is a two-stage procedure, so its error rates are only approximately nominal. The fitness shapes the gait (Section 2.3), but the gait terms add only 0.18–0.20 to the best controllers’ fitness, and the plain distance gives the same ordering (Table 3). The 15 s walk cuts off slow but working gaits, and the runs had not plateaued (after 9,000 evaluations, 12–16 of each EA’s 20 runs still improved). Finally, we used one body and one world, and the controllers specialise to their arena; training on varied arenas [13] did not help in a pilot.

## 6 CONCLUSION

We asked whom an island model should send between its islands when evolving neural controllers that make spider\_8 walk to a target. With 20 paired seeds, every EA clearly outperforms random search, but the emigrant-selection policy matters little: migrating the best or random individuals probably speeds up evolution compared with no migration, and no difference between two EAs is significant after correction. Migrating the best made the islands as alike as one population, as the literature predicts, at little cost within 166 generations; migrating the worst behaved like no migration. On our data, random migration is a safe default: about as fast as migrating the best, it ended fittest and was never the worst on any seed. These conclusions rest on one body, one world, a shaped fitness and effects too small to confirm with 20 seeds. Future work: longer budgets, where island diversity should pay off; crossover that aligns immigrant and native neurons [22]; and more varied training arenas.

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## CONTRIBUTIONS

[TODO: each member: one or two lines on what they did]

## USE OF GENERATIVE AI

We used Anthropic’s Claude, through Claude Code, as an assistant throughout the project: to write and review code (the EA, the simulation and fitness, the analysis and figure scripts and their tests), to propose design options and hypotheses, to run and monitor the experiments, and to draft and edit the text of this report. The team chose the research question, made the design decisions and reviewed all code and text. Every number in this report comes from our own runs and was checked against their logs; we take full responsibility for the content.